

Continuous functions – 2

Monotonicity, oscillation and the horizontal chord

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Theoretical minimum.

Theorem 1 (horizontal chord). If f is continuous on $[0, 1]$ and $f(0) = f(1)$, then for every $n \in \mathbb{N}$ there is x with $f(x + \frac{1}{n}) = f(x)$. For a length α which is not of the form $\frac{1}{n}$ there is a continuous f with $f(0) = f(1)$ and $f(x + \alpha) \neq f(x)$ for all x .

Theorem 2 (monotonicity and range). A continuous injective function on an interval is strictly monotone, and its inverse is continuous. A monotone function has at most countably many discontinuities, all of them jumps; the range of a monotone function is an interval if and only if the function is continuous.

Fact 1. The trick of the sheet: replace f by the difference $g(x) = f(x + \alpha) - f(x)$ or by $f(x) - x$, and apply the intermediate value theorem to g . Whenever a statement claims the existence of a point with a prescribed property, this is the first thing to try.

1. Prove that there is no increasing function defined on all of $\mathbb{R}$ whose set of values is exactly the set of all irrational numbers.
2. Is there a strictly increasing function $f: \mathbb{R} \rightarrow \mathbb{R}$ taking only irrational values?
3. For which $\alpha \in (0, 1)$ can one assert the following: if $f: [0, 1] \rightarrow \mathbb{R}$ is continuous and $f(0) = f(1)$, then there exists $x \in [0, 1 - \alpha]$ with $f(x + \alpha) = f(x)$?
4. Let $f: (0, +\infty) \rightarrow \mathbb{R}$ be bounded and continuous. Prove that for every $T > 0$ there is a sequence $x_n \rightarrow \infty$ with $f(x_n + T) - f(x_n) \rightarrow 0$.

5. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfy $\lim_{x \rightarrow x_0} f(x) = 0$ for every $x_0 \in \mathbb{R}$. Prove that there exists x_0 with $f(x_0) = 0$.